UART

→ Universal Asynchronous Receiver & Transmitter

* UART is much much slower than USB but its simple & many peripherals or sensors actually use it.

* both of them do not share any common clock, they just agree upon no. of bits, parity . . .

Sampling = 16 × Baude rate

→ rate at which it is pooling

Oversampling

1. wait until incoming signal becomes 0 (

start bit), then start the sampling tick counter.

2. When tick counter reaches 7 (middle of start bit), clear tick counter & reset.

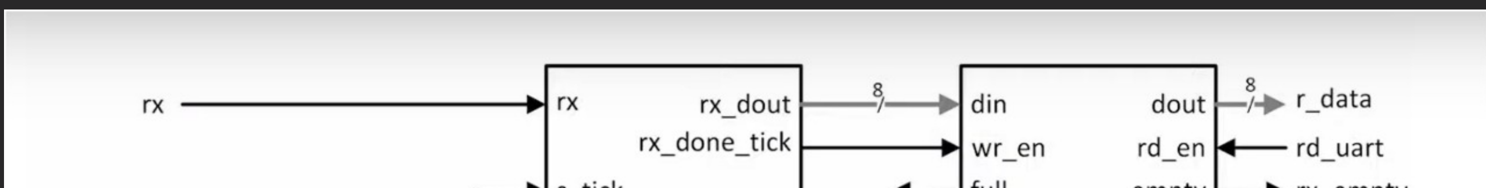
3. when counter reaches 15 (middle of first data bit), shift bit value into register & reset tick counter.

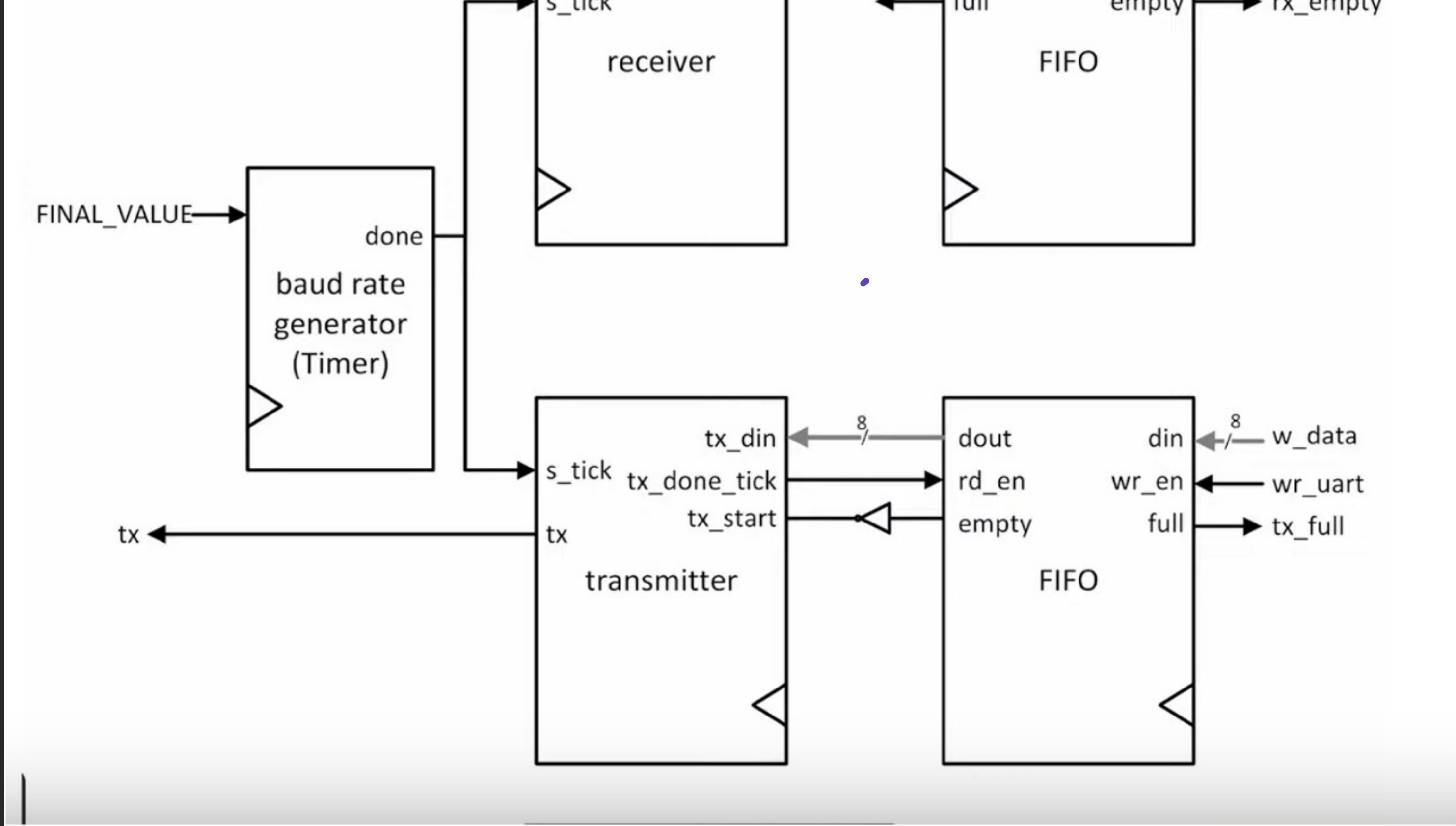
4. Repeat step.3 (N-1) times to retrieve the remaining data bits.

5. If optional parity bit is used, repeat step 3. one time more.

6. Repeat step 3

UART Construction





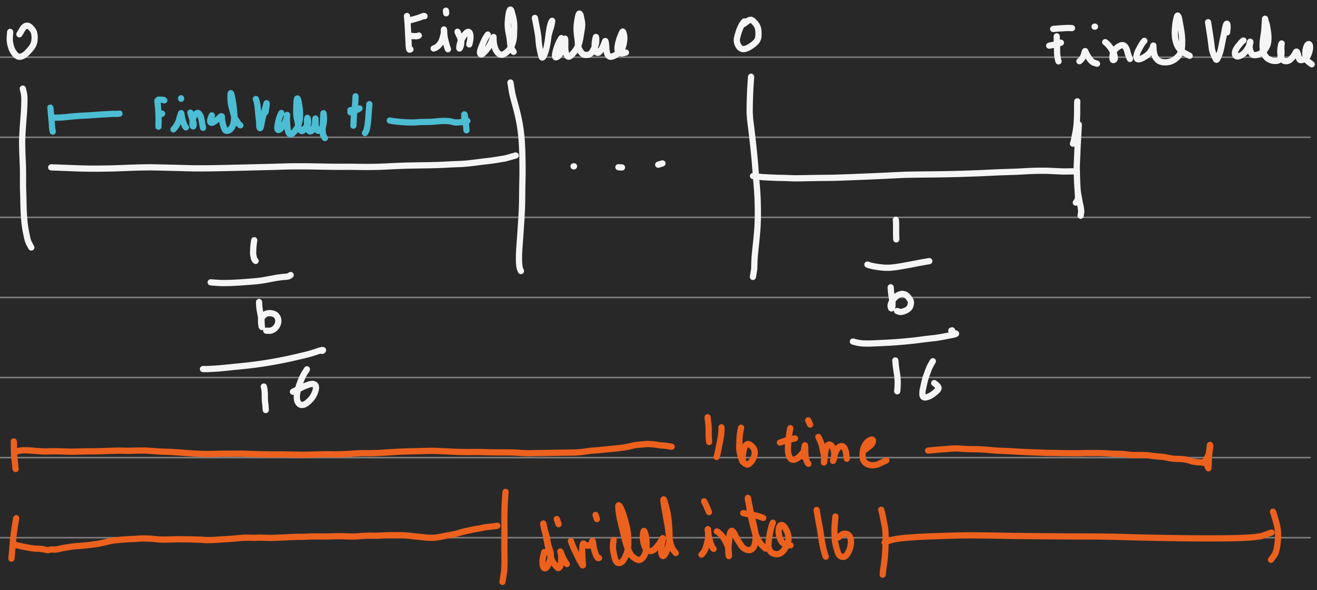
@ Baud rate generator: it just ticks every certain period of time based on baud rate.

- ticks 16 x baud rate.
- it's like a timer

@ Receiver is receiving at UART rate which is slower than transmitter rate which is connected to Processor

Baud Rate Generator

↳ Calculations



system clock cycle

$$(\text{Final Value} + 1) \times T = \frac{1}{16 \times b}$$

$$\Rightarrow \text{Final Value} = \frac{1}{16 \times b \times T} - 1$$

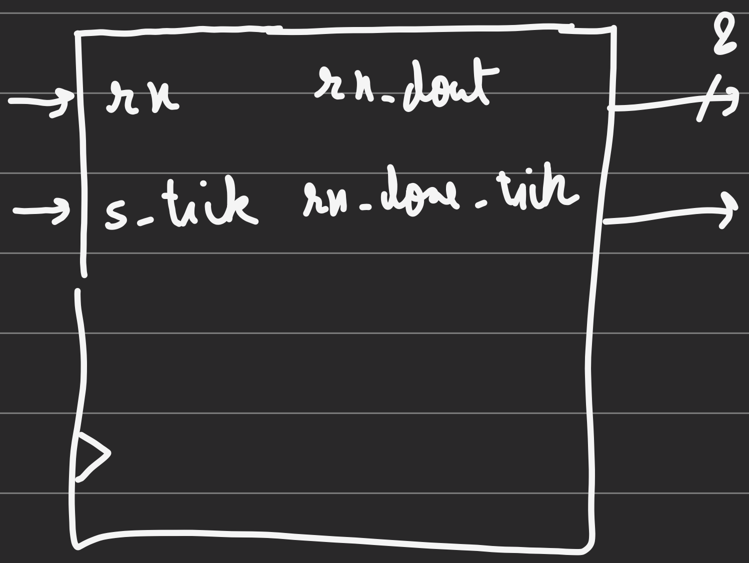
rate at which we transmit the data.

<u>Ex</u> $f = 100\,MHz$, $b = 9600$ bits/sec.

then $$\text{Final value} = \frac{\cancel{100} \times 10^6}{16 \times 96\cancel{00} \times 2} - 1$$

$$\approx 650$$

Receiver



data

ASMD